H2DF: Calibrated Target Bands for Controllable Language-Model Unlearning

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Abstract

Gradient ascent (GA) is a simple language-model unlearning baseline, but it continues increasing forget loss after useful forgetting has occurred. This unbounded pressure can damage retained utility and make forgotten examples abnormally high-loss. We introduce H2DF, a calibrated target-band objective that applies GA-like pressure below a data-dependent lower target and suppresses, or reverses, that pressure after the target region is reached. The targets are estimated from matched non-member examples before training, so the base objective requires one forget-batch forward/backward pass per optimization step; our reported TOFU variant adds sparse retain replay every tenth step. We instantiate a one-sided objective for TOFU question answering and a two-sided objective for MUSE-News language modeling. Across three TOFU seeds, H2DF obtains a forget NLL of 1.528 ± 0.049 and retain NLL of 1.165 ± 0.047, matching GradDiff while using retain replay only every tenth step and showing substantially lower variability than GA. On MUSE-News, H2DF improves retention over GA and NPO and is competitive with SimNPO, but does not dominate it. Ablations show that the upper target reduces overshoot and that validation-based checkpoint selection is important. These results support calibrated target bands as a practical mechanism for controlling forgetting strength, while also exposing sensitivity to random seed and calibration quality.

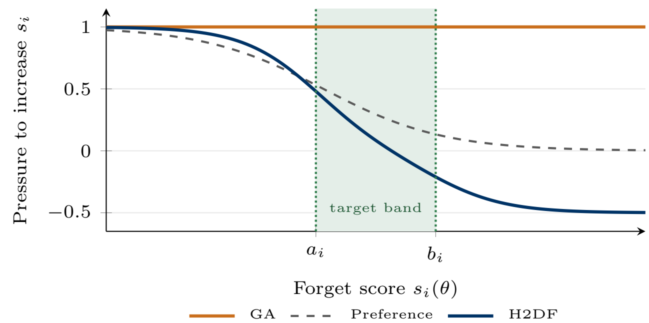

Figure 1: Conceptual forget pressure. GA remains persistent; preference-style objectives attenuate pressure implicitly [6, 22]; H2DF pushes below a_i, saturates in the calibrated band, and can restore above b_i. Curves use illustrative parameters.

1 Introduction

Large language models may memorize private, copyrighted, or otherwise undesirable training data. Machine unlearning seeks to remove the influence of selected data without retraining from scratch [2]. For large models, a common baseline is gradient ascent on the forget set [21]. Given a forget batch B_f and per-example negative log-likelihood (NLL) $\ell_i(\theta)$, GA minimizes

$$\mathcal{L}_{\mathrm{GA}}(\theta) = -\frac{1}{|B_f|}\sum_{i\in B_f}\ell_i(\theta). \tag{1}$$

GA is inexpensive, but it treats forgetting as an infinite direction: the objective continues to increase NLL even after the model no longer reproduces the targeted content. This can cause catastrophic utility loss and abnormal high-loss artifacts. Preference-based methods such as NPO and SimNPO temper this behavior [6, 22], while retain-aware methods such as GradDiff add utility-preserving supervision at additional online cost. Figure 1 summarizes the central distinction.

Figure 2 gives a selective view of the objective-design path that motivates our method. Classical work focused on deleting training influence or approximating retraining [1, 2, 7]. LLM unlearning made direct forget-loss maximization a scalable baseline [21]; retain-aware and preference objectives then sought to limit utility collapse

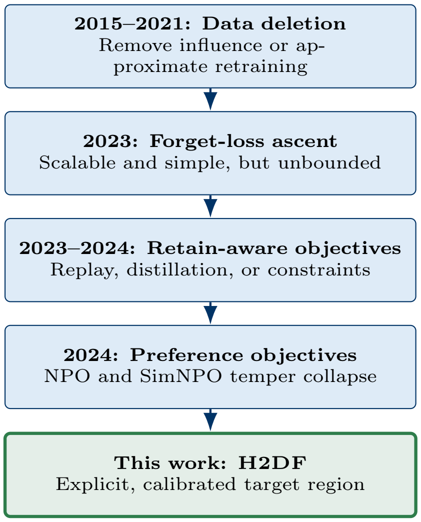

Figure 2: Selective progression of unlearning objectives. H2DF makes the stopping region explicit rather than pushing indefinitely away from the original likelihood.

[5, 6, 22]. H2DF continues this progression by making the stopping region explicit and calibrating it from non-members.

We study a direct alternative: forgetting should be a calibrated region rather than an unbounded direction. H2DF estimates a target loss interval from matched non-member data and optimizes each forget example toward that interval. Below the interval, its gradient follows GA; inside the interval, pressure saturates; above the interval, an optional upper penalty discourages overshoot. Calibration is performed once, and training uses only the current model's forget-batch losses.

Design principle. *"Forgetting should be a calibrated region, not an infinite direction."*

Our contributions are:

(i) a general target-band objective that recovers GA-like updates when forgetting is insufficient and suppresses them after a calibrated target is reached;
(ii) task-specific instantiations for TOFU [14] and MUSE [18];
(iii) experiments against GA, GradDiff, NPO, and SimNPO, including three-seed TOFU results, MUSE-News official metrics, and target-band ablations; and
(iv) an empirical analysis of overshoot, checkpoint selection, and seed sensitivity in short-budget LoRA unlearning.

2 H2DF Method

2.1 General Objective

Let $s_i(\theta)$ be a score for forget example i, with larger values denoting stronger forgetting. In our experiments, s_i is token-averaged NLL. Let $[a_i, b_i]$ be a target interval estimated from a held-out non-member calibration set, conditional on observable features c_i such as sequence length, domain, and original-model loss. We define

$$a_i = Q_{\rho_{\text{low}}}(s_{\text{cal}} \mid c_i), \qquad b_i = Q_{\rho_{\text{high}}}(s_{\text{cal}} \mid c_i), \tag{2}$$

where Q_ρ is an empirical quantile. Figure 3 summarizes how these fixed targets alter the online forget update.

Using the smooth hinge $h_\tau(u) = \tau \log(1 + \exp(u/\tau))$, the objective is

$$\mathcal{L}_{\text{H2DF}}(\theta) = \frac{1}{|B_f|} \sum_{i \in B_f} \left[h_\tau(a_i - s_i(\theta)) + \gamma h_\tau(s_i(\theta) - b_i) \right] + \lambda \|\theta_{\text{LoRA}}\|_2^2. \tag{3}$$

The lower term penalizes insufficient forgetting. The upper term, weighted by $\gamma \geq 0$, penalizes excessive loss; $\gamma = 0$ gives a one-sided objective. The targets are precomputed, so Eq. (3) has the same online forward/backward structure as GA.

2.2 Gradient Behavior

For the one-sided objective,

$$\begin{aligned} \nabla_\theta h_\tau(a_i - s_i(\theta)) &= -\alpha_i(\theta) \nabla_\theta s_i(\theta), \\ \alpha_i(\theta) &= \sigma\left(\frac{a_i - s_i(\theta)}{\tau}\right). \end{aligned} \tag{4}$$

Thus $\alpha_i \approx 1$ when $s_i \ll a_i$, recovering the direction of GA, and $\alpha_i \to 0$ after s_i exceeds the lower target. With an upper target,

$$\frac{\partial \mathcal{L}_i}{\partial s_i} = -\sigma\left(\frac{a_i - s_i}{\tau}\right) + \gamma\sigma\left(\frac{s_i - b_i}{\tau}\right), \tag{5}$$

so sufficiently large losses receive a restoring gradient. This bounded coefficient does not bound the model Jacobian $\nabla_\theta s_i$; the claim is specifically that the objective removes GA's persistent scalar incentive to increase loss.

2.3 Task Instantiations

TOFU. For question q_i and answer tokens $y_{i,1:T_i}$, the score is answer-only NLL:

$$s_i(\theta) = -\frac{1}{T_i} \sum_{t=1}^{T_i} \log p_\theta(y_{i,t} \mid q_i, y_{i,<t}). \tag{6}$$

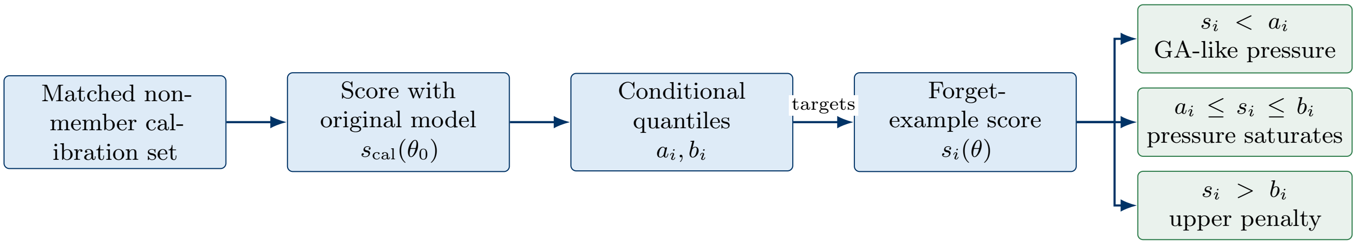

Figure 3: H2DF workflow. Calibration converts matched non-member scores into per-example target regions before training. During unlearning, examples below the lower target receive GA-like pressure, examples inside the target region receive little pressure, and the optional upper term discourages overshoot.

We use a one-sided target

$$a_i = \max\{s_i(\theta_0) + \Delta_{\min}, Q_\rho(s_{\mathrm{cal}}(\theta_0) \mid c_i)\}, \qquad (7)$$

which combines a minimum displacement from the original model with a non-member quantile.

MUSE-News. For text $x_{i,1:T_i}$, s_i is token-averaged language-modeling NLL. We use the two-sided targets in Eq. (2). The goal is not to maximize perplexity, but to move forgotten text toward the loss regime of matched non-members.

2.4 Illustrative Unlearning Example

Figure 4 illustrates the mechanism on a synthetic TOFU-style question. Suppose the original model assigns answer NLL 0.4 to a memorized answer, while matched non-member answers define a target band $[1.4, 2.2]$. Both GA and H2DF initially increase the answer NLL. After H2DF enters the band, its lower-target pressure vanishes and the upper term prevents large overshoot. GA has no corresponding stopping signal and continues increasing loss. The numerical trajectory is illustrative rather than an experimental trace; it visualizes the gradient behavior in Eq. (4).

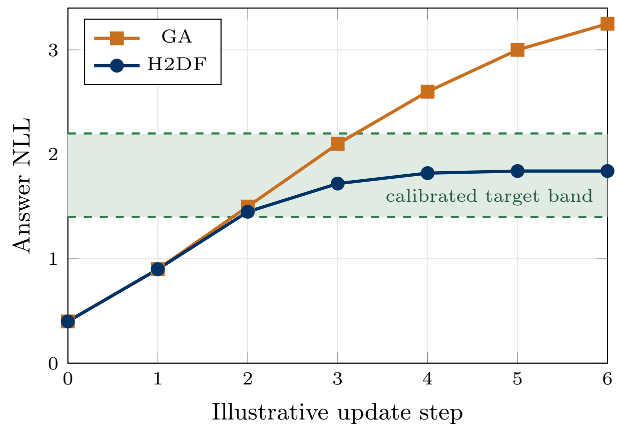

Figure 4: Illustrative per-example dynamics. For a synthetic memorized answer with target band $[1.4, 2.2]$, GA continues increasing NLL, whereas H2DF approaches the non-member-like region and saturates. Values are pedagogical, not measured experiment outputs.

3 Experimental Setup

TOFU. We use the TOFU forget-5% split and the remaining authors for retention and calibration. The base is `Qwen2.5-1.5B-Instruct` with a supervised TOFU adapter, followed by rank-8 LoRA unlearning. Runs use AdamW, learning rate 10^{-4}, batch size 4, 500 steps, cosine scheduling, $\tau = 0.25$, $\rho = 0.7$, $\Delta_{\min} = 0.5$, and $\lambda = 10^{-4}$. We report seeds 42, 43, and 44 where available. Evaluation covers 200 forget and 3,420 retain examples; exact match is close to zero for all methods, so token F1 is more discriminative. The reported H2DF variant applies a retain batch every 10 steps; GradDiff applies retain supervision every step.

MUSE-News. For MUSE-News, we use `muse-bench/MUSE-News_target` as the target model/checkpoint and the `NousResearch/Llama-2-7b-hf` tokenizer. Rank-8 LoRA runs use AdamW, learning rate 2×10^{-5}, effective batch size 4, 50 steps, $\tau = 0.25$, $\gamma = 0.5$, $(\rho_{\text{low}}, \rho_{\text{high}}) =$

$(0.5, 0.9)$, and $\lambda = 10^{-4}$. Full internal evaluation contains 888 forget, 1,778 retain, and 1,521 calibration examples.

Baselines and selection. We compare with the original model, GA, GradDiff, NPO, and SimNPO. All compared unlearning methods use the same LoRA parameterization. For MUSE, checkpoints are screened on a fixed 200-example validation subset and selected by proximity to 50% of examples below the lower target. The screen is used only for checkpoint selection; reported seed-42 internal results use the full held-out splits. Official MUSE metrics are reported for the original model, H2DF, and SimNPO. Appendix A provides the training algorithm, exact configurations, evidence provenance, and checkpoint-selection rule.

Metrics. For TOFU, higher forget NLL and lower forget token F1 indicate stronger forgetting; lower retain NLL and higher retain token F1 indicate better utility. For MUSE internal diagnostics, we report forget and retain NLL, loss-based membership-inference AUC, and the fraction of forget examples inside the calibrated band. AUC is diagnostic only and does not replace the official MUSE suite. Official VerbMem-Forget and KnowMem-Forget are lower-is-better; KnowMem-Retain is higher-is-better. We reproduce PrivLeak with the official evaluator and preserve its sign convention.

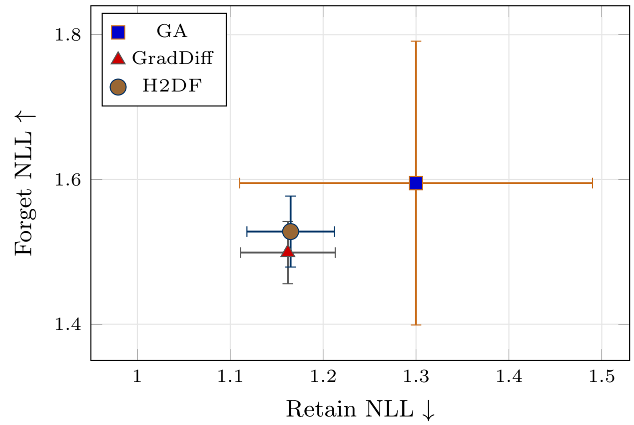

Figure 5: TOFU forgetting–retention tradeoff over seeds 42–44. Points are means and error bars are sample standard deviations. The preferred region is toward the upper left. H2DF and GradDiff are stable and closely matched, whereas GA has substantially larger variance.

Table 1: TOFU seed-42 results. Arrows indicate the preferred direction; best unlearning values are shaded.

	Forget		Retain	
Method	NLL↑	F1↓	NLL↓	F1↑
Original/SFT	1.004	0.441	0.997	0.446
GA	1.560	0.312	1.280	0.334
NPO	1.561	0.310	1.282	0.333
SimNPO	**1.643**	**0.287**	1.365	0.309
GradDiff	1.538	0.318	1.220	0.353
H2DF	1.565	0.327	**1.217**	**0.360**

Table 2: TOFU mean ± sample standard deviation over seeds 42–44.

	Forget		Retain	
Method	NLL↑	F1↓	NLL↓	F1↑
GA	1.595±0.196	**0.310±0.042**	1.300±0.190	0.341±0.054
GradDiff	1.499±0.043	0.348±0.026	**1.162±0.051**	0.389±0.032
H2DF	**1.528±0.049**	0.353±0.022	1.165±0.047	**0.394±0.029**

4 Results

4.1 TOFU

Table 1 shows the seed-42 tradeoff. SimNPO forgets most strongly but also incurs the largest utility loss. H2DF and GradDiff occupy a less aggressive region: H2DF has slightly higher forget NLL and higher retained token F1 than GradDiff, while their NLLs are nearly identical. GA and NPO are also nearly identical in this run.

Across three seeds (Table 2 and Fig. 5), H2DF and GradDiff remain close. H2DF achieves somewhat stronger NLL forgetting and slightly higher retain F1, whereas GradDiff has lower forget F1 and marginally lower retain NLL. GA is more aggressive on average but substantially more variable; its seed-44 run raises both forget and retain NLL. The result supports competitive forgetting–retention control, not uniform dominance.

4.2 MUSE-News

Table 3 reports the full held-out loss diagnostics. GA and NPO attain the lowest loss-based MIA AUC, but at the cost of the largest retain degradation. H2DF reduces retain NLL by 0.35 relative to GA and NPO while maintaining a lower AUC than GradDiff. SimNPO is the closest comparator: it has better retain NLL and a higher in-band rate, while H2DF has slightly lower AUC and higher forget NLL. These diagnostics therefore show a competitive, not dominant, tradeoff.

Official metrics in Table 4 show that H2DF and SimNPO are close on knowledge forgetting. SimNPO has stronger verbatim forgetting, while H2DF retains slightly more knowledge utility. We report PrivLeak using the official evaluator's sign convention and do not interpret its direction here. Official MUSE evaluation was completed only for seed 42, so these differences should not be interpreted as statistically stable.

Table 3: MUSE-News seed-42 held-out diagnostics. MIA AUC is diagnostic; Band is the fraction inside the calibrated target interval.

Method	Forget NLL	Retain NLL↓	MIA AUC↓	Band fraction↑
Original	0.576	0.761	0.981	0.015
GA	1.865	2.018	**0.934**	0.071
NPO	1.858	2.010	**0.934**	0.072
GradDiff	1.379	**1.546**	0.961	0.081
SimNPO	1.513	1.597	0.955	**0.144**
H2DF	1.557	1.666	0.949	0.131

Table 4: Official MUSE-News seed-42 metrics. PrivLeak follows the evaluator's sign convention and is not direction-ranked.

	Forget↓		Retain↑	
Model	VerbMem	KnowMem	KnowMem	PrivLeak
Original	57.42	64.09	54.12	-99.81
SimNPO	**19.27**	**45.74**	38.27	-98.32
H2DF	20.71	45.83	**38.72**	-96.52

4.3 Target-Band Ablations

The upper target reduces overshoot (Table 5 and Fig. 6): at step 40, the two-sided objective places 30.0% of examples above the band, compared with 47.0% for the lower-only variant, and improves retain NLL from 1.790 to 1.598. Continuing to the fixed final checkpoint substantially increases the above-band fraction and retention damage for both objectives. Sparse retain replay with weight 0.1 moves in the opposite direction and leaves 90.5% of examples below target. These results motivate target-aware checkpoint selection rather than always using the final training step.

The corrected MUSE multiseed screen selects steps 40, 50, and 50 for seeds 42, 43, and 44, respectively. Their forget NLLs are 1.530, 0.956, and 1.531. Seed 43 leaves 93.0% of examples below target, showing that a fixed 50-step budget does not reliably reach the intended region.

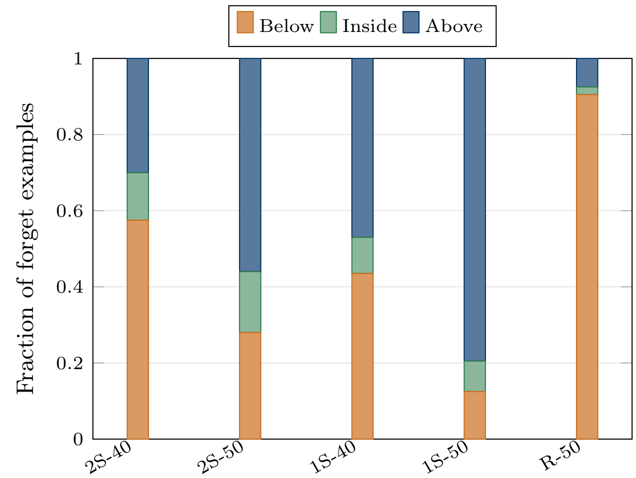

Figure 6: MUSE validation-screen target occupancy. 2S and 1S denote two-sided and lower-only objectives; the suffix is the checkpoint step, and +R denotes sparse retain replay. The upper target reduces overshoot at step 40, while fixed final checkpoints overshoot and retain replay under-forgets.

Table 5: MUSE validation-screen ablations. Occupancy columns are fractions relative to the calibrated band.

Variant	Step	Forget NLL	Retain NLL	Below	Inside	Above
Two-sided	40	1.530	1.598	0.575	0.125	0.300
Two-sided final	50	1.800	1.852	0.280	0.160	0.560
Lower-only	40	1.711	1.790	0.435	0.095	0.470
Lower-only final	50	2.107	2.171	0.125	0.080	0.795
Two-sided+R	50	1.107	1.195	0.905	0.020	0.075

5 Analysis and Discussion

Computational cost. Let $\mathcal{F}(b)$ denote the cost of a forward pass on batch size b, with backward cost approximately $2\mathcal{F}(b)$. GA, SimNPO, and H2DF each require one current-model forget forward/backward pass, approximately $3\mathcal{F}(|B_f|)$ per step. NPO additionally requires stored original losses or a reference computation, depending on implementation. GradDiff adds a retain forward/backward pass every step. H2DF shifts its additional work to one-time target calibration.

What the target band controls. The objective directly controls per-example loss relative to a non-member reference distribution. It does not certify removal of training-data influence, nor does matching losses guarantee matching all behavioral or privacy properties. The official MUSE results illustrate this distinction: internal loss diagnostics are useful for selection, but do not determine VerbMem, KnowMem, or PrivLeak.

Calibration uncertainty. For n calibration examples in a conditioning bin, the Dvoretzky–Kiefer–Wolfowitz inequality gives, with probability at least $1 - \delta$,

$$\sup_z |\widehat{F}_n(z) - F(z)| \le \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{8}$$

This motivates coarse bins and minimum bin sizes. The bound addresses sampling error, not distribution mismatch: a large but mismatched calibration set can still define the wrong target.

Practical implication. The experiments suggest that the upper target and checkpoint rule are as important as the lower forgetting target. A lower-only objective can still overshoot, and retain replay can prevent target attainment if weighted too strongly. A useful deployment protocol should therefore report target occupancy through training and validate checkpoint selection on held-out data.

6 Related Work

Classical machine unlearning. Machine unlearning was originally motivated by the need to remove selected training examples without retraining a model from scratch [2]. Early work studied both exact and approximate deletion mechanisms. SISA training reduces the cost of deletion by partitioning training into shards and slices [1], while certified-removal methods provide stronger guarantees in restricted model classes [7, 9, 16]. Other approaches modify, reverse, or approximate training updates to remove the influence of targeted examples [8, 10, 19]. These works establish the central tension between deletion quality, computational cost, and retained utility, but they do not directly solve unlearning for modern generative language models.

Unlearning for large language models. LLM unlearning is harder because the forget target is often semantic, generative, and difficult to localize. Gradient-ascent-style objectives are simple and widely used, but they can cause utility collapse or abnormal generations [21, 22]. Recent methods therefore add retain losses, distillation, preference-style objectives, or optimization constraints to better trade off forgetting and utility [5, 12, 17, 20]. Our method follows this line of work but focuses on a different control mechanism: rather than maximizing forget loss indefinitely, it defines a data-calibrated target region for each forget example.

Preference- and representation-based methods. Negative Preference Optimization reframes unlearning as a preference-style objective and slows catastrophic collapse compared with GA [22]. SimNPO simplifies this formulation by removing the reference-model dependence and obtains strong empirical results on TOFU and MUSE [6]. Representation-based methods such as RMU steer internal activations away from forget-domain representations and are especially important for hazardous-knowledge removal [11]. Follow-up work shows that apparent forgetting can fail under stronger robustness and knowledge-extraction tests [13]. H2DF is complementary: it does not directly steer hidden states or use preference pairs, but instead calibrates the loss region that forgotten examples should reach.

Benchmarks and evaluation. TOFU provides a controlled fictitious-author QA benchmark for testing whether a fine-tuned model behaves as if selected author profiles had not been learned [14]. MUSE evaluates unlearning across verbatim memorization, knowledge memorization, privacy leakage, utility, scalability, and sustainability [18]. WMDP evaluates hazardous-knowledge removal and introduced representation misdirection as a strong unlearning baseline [11]. Other evaluations study membership inference, extraction, and memorization risks in language models [3, 4, 15]. These benchmarks show that high forget loss alone is not sufficient: unlearning methods must also preserve retain behavior and avoid new privacy or utility failures. This motivates H2DF's calibrated target-band view.

7 Limitations

The study has four main limitations. First, MUSE official metrics are available only for seed 42, and the validation screens reveal substantial seed sensitivity. Second, MUSE-Books, sequential requests, and larger-scale models are not evaluated. Third, TOFU and MUSE use different base models and evaluation regimes, so their numerical results are not directly comparable. Fourth, the loss-based target is a proxy rather than a certified unlearning guarantee; performance depends on the quality and representativeness of the calibration set. The released RMU implementation uses LoRA rather than the original full-parameter setup and is therefore excluded from the main quantitative comparison.

8 Conclusion

We introduced H2DF, a calibrated target-band formulation for language-model unlearning. It preserves GA's early update direction while suppressing pressure after examples reach a non-member-derived loss region. On TOFU, H2DF matches the forgetting–retention tradeoff of GradDiff across three seeds with low variability. On

MUSE-News, it improves retention over GA and NPO and is competitive with, but does not outperform, SimNPO. The upper target reduces overshoot, while multiseed results show that checkpoint selection and optimization stability remain open practical problems. Calibrated target bands are therefore best viewed as a mechanism for controllable forgetting, not as a guarantee of complete data removal.

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A Training Algorithm

Algorithm 1 Calibrated target-band unlearning. Calibration is completed before optimization; targets are not re-estimated from evaluation data.

Require: Original model θ_0; forget set D_f; matched calibration set D_{cal}; optional retain set D_r; quantiles $\rho_{\text{low}}, \rho_{\text{high}}$; τ, γ, λ; replay period K; training steps T

Ensure: LoRA adapter, fixed targets, diagnostics, and selected checkpoint

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Score D_f and D_cal with θ_0; compute length, original-loss, and available domain/type features
Build coarse feature bins; progressively back off conditioning features when a matched bin is smaller than the minimum bin size
for each forget example i do
    Compute a_i and optional b_i from matched calibration quantiles
    if task is TOFU then
        a_i ← max{a_i, s_i(θ_0) + Δ_min}
    end if
end for
Initialize trainable LoRA parameters θ
for t = 1, ..., T do
    Sample B_f ⊂ D_f and compute per-example scores s_i(θ)
    L ← L_H2DF(θ; B_f) from Eq. (3)
    if D_r is used and t mod K = 0 then
        L ← L + β_r L_retain(θ; B_r)
    end if
    Backpropagate, clip gradients, update θ, and log target occupancy
end for
Select a saved checkpoint on the fixed validation screen; evaluate once on held-out data
```

B Reproducibility Details

Table 6: Primary configurations used for the reported experiments.

Setting	TOFU	MUSE-News
Base model/checkpoint	Qwen2.5-1.5B-Instruct with TOFU SFT adapter	MUSE-News target checkpoint
Tokenizer	model tokenizer	Llama-2-7B tokenizer
LoRA rank / alpha / dropout	8/16/0	8/16/0
Target modules	q, k, v, o, gate, up, and down projections	
Precision	BF16	BF16
Batch / accumulation	4/1	1/4
Maximum sequence length	512	1024
Optimizer / schedule	AdamW / cosine	AdamW / cosine
Learning rate / steps	$10^{-4}/500$	$2 \times 10^{-5}/50$
Warmup / gradient clipping	25/1.0	25/1.0
τ / λ	$0.25/10^{-4}$	$0.25/10^{-4}$
Calibration quantiles	$\rho = 0.7$, $\Delta_{\min} = 0.5$	$(\rho_{\text{low}}, \rho_{\text{high}}) = (0.5, 0.9)$
Upper-band weight γ	0	0.5
Retain replay	every 10 steps for reported H2DF variant	none for the main H2DF run

Table 7: Evidence provenance. "Screen" denotes the fixed 200-example checkpoint-diagnostic subset rather than final held-out evaluation; * indicates availability for only a subset of methods.

Evidence block	Seeds	Evaluation scope	Manuscript use
TOFU full QA	42–44*	200 forget; 3,420 retain examples	Tables 1–2, Fig. 5
MUSE internal comparison	42	888 forget; 1,778 retain; 1,521 calibration	Table 3
MUSE official suite	42	Official VerbMem, KnowMem, and PrivLeak evaluation	Table 4
MUSE multiseed screen	42–44	Fixed 200-example validation screen	Seed-sensitivity discussion
MUSE ablation screen	42	Fixed 200-example validation screen	Table 5, Fig. 6

C Checkpoint Selection and Reporting

For the short-budget MUSE runs, let

$$r_t = \frac{1}{|V_f|} \sum_{i \in V_f} \mathbf{1}[s_i(\theta_t) < a_i] \tag{9}$$

be the fraction of validation forget examples still below their lower target at checkpoint t. We selected

$$t^\star = \arg\min_{t \in \mathcal{T}} |r_t - 0.5|, \tag{10}$$

using a fixed 200-example validation screen and the saved checkpoint set $\mathcal{T}$. This criterion was chosen before full held-out evaluation to avoid selecting directly on the reported test metrics. It is a practical heuristic rather than a theoretically optimal rule. The manuscript therefore reports target occupancy alongside final benchmark metrics and distinguishes validation-screen ablations from full held-out and official evaluations.

D Artifact Contents

The released artifact contains the training and calibration implementation, benchmark-specific YAML configurations, baseline runners, CPU unit tests, and the compact JSON metrics used in the manuscript. Model weights, adapters, downloaded datasets, credentials, and raw experiment caches are intentionally excluded. The expected workflow is:

1. run calibration to create fixed per-example targets;
2. train H2DF or a configured baseline;
3. evaluate the selected adapter on forget, retain, and calibration data;
4. run the official MUSE evaluator for benchmark-level metrics.